

Semester: July 2025 – November 2025		
Maximum Marks:30	Examination: In-Semester Examination (Mid Semester Examination)	Duration :1.15
Programme code: 54 Programme: Honours in Data Science and Analytics	Class: LY	Semester: VII (SVU 2020)
Name of the School: K. J. Somaiya School of Engineering	Name of the department: COMP/EXCP	
Course Code: 116h54C701	Name of the Course: Advanced Machine Learning	

Question No.		Max. Marks
Q1	a) Differentiate between overfitting and underfitting with the help of diagrams and examples. (5M) b) Explain the role of Cross Validation in Machine Learning. How does it help in improving model performance? (5M) OR a) Define Machine Learning. Explain the three types of machine learning with suitable examples and write short notes on Bias and Variance in Machine Learning.	10
Q2	Consider a neural network with the following architecture: • Input layer: 3 neurons • Hidden Layer 1: 4 neurons with ReLU activation • Hidden Layer 2: 3 neurons with Sigmoid activation • Output Layer: 1 neuron with Sigmoid activation For the input vector: $x=[0.6, 0.2, 0.9]$ The target output is: $y=1$. Assume that after forward propagation, the network produces the predicted output: $=0.72$ a) Compute the binary cross-entropy loss for this prediction. (4 M) b) Explain how backpropagation and optimization will update the weights to reduce this loss in subsequent iterations. (6 M)	10
Q3	Draw and describe the architecture of the Convolutional Neural Network - VGG-19. In your explanation, address the following points: 1. Specify the total number of layers in the VGG-19 model. 2. Identify and describe the types of layers used (e.g., convolutional, max pooling, fully connected). 3. Mention the activation functions applied throughout the network. 4. Calculate and provide the output dimensions after each layer, particularly after every convolutional and pooling operation.	10